Retail Customer Segmentation and Pattern Discovery Using Data Mining Techniques

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***Abstract*—Understanding customer purchasing behavior is an important task in modern online retail businesses. This project applies multiple data mining techniques to analyze customer behavior using the UCI Online Retail dataset. Association rule mining, customer segmentation, PCA visualization, decision tree interpretation, Naive Bayes analysis, and anomaly detection were applied to discover meaningful customer purchasing patterns. The results identified VIP customers, loyal customer groups, inactive customers, and unusual purchasing behaviors. The project also demonstrates how combining multiple analytical approaches can provide useful business insights for recommendation systems, customer retention, and marketing strategies.**

I. Introduction

This project uses the Online Retail dataset from the UCI Machine Learning Repository. The dataset contains 541,909 transaction records from a UK-based online retailer between 2010 and 2011. Each transaction includes information such as product description, quantity, invoice date, customer ID, and unit price.

The main goal of this project is to discover meaningful customer purchasing patterns using multiple data mining techniques. The project focuses on three main discovery questions:

1) Which products are frequently purchased together?
2) Are there natural customer segments based on purchasing behavior?
3) Are there anomalous customers or unusual purchasing behaviors?

To answer these questions, several data mining techniques were applied throughout the project. Association rule mining using Apriori was used to identify frequently co-purchased products. Customer segmentation was performed using RFM features and K-Means clustering. PCA was used to improve visualization and cluster separation. Decision trees were applied to better interpret customer clusters, while Naive Bayes was used to analyze probabilistic relationships between RFM features and customer groups. Finally, Isolation Forest and Local Outlier Factor (LOF) were used for anomaly detection.

The results reveal several important customer behavior patterns. Some customers spend significantly more money and purchase much more frequently than others, while some customers appear inactive or at risk of churn. The analysis also shows that many unusual customers overlap with VIP customer groups, suggesting that high-value customers often behave very differently from average customers.

Overall, this project demonstrates how different data mining techniques can work together to better understand customer behavior and generate useful business insights for online retail companies.

II. Methods

A. Data Preprocessing and Feature Engineering

Before applying the data mining techniques, several preprocessing steps were performed to improve the quality of the dataset. Records with missing CustomerID values were removed because customer-level analysis requires valid customer identifiers. Transactions with negative quantities were also removed since they represent product returns or cancellations rather than actual purchases. In addition, rows with zero or negative unit prices were filtered out to avoid unrealistic transaction values.

To represent customer purchasing behavior, RFM features were constructed:

- **Recency**: how recently a customer made a purchase
- **Frequency**: how often a customer made purchases
- **Monetary**: the total amount spent by the customer

These RFM features provide a simple but effective representation of customer value and engagement.

B. Data Mining Techniques

1) Association Rule Mining: Apriori association rule mining was used to identify products that are frequently purchased together. Association rules were generated and evaluated using support, confidence, and lift metrics to identify strong product relationships.

2) Customer Segmentation: Customer segmentation was performed using K-Means clustering based on the RFM features. K-Means was selected because RFM data is numerical and easy to interpret using cluster centroids.The number of clusters was set to four based on interpretability of the RFM customer segments.

3) PCA Visualization: Principal Component Analysis (PCA) was applied to improve visualization and cluster separation. PCA reduced feature correlation and transformed customer behavior into lower-dimensional space.

4) Decision Tree Interpretation: Decision trees were used to interpret clustering results rather than for prediction purposes. The goal was to identify interpretable business rules. The decision tree depth was limited to improve interpretability and reduce overfitting.

5) Naive Bayes Analysis: Naive Bayes was applied to analyze probabilistic relationships between RFM features and customer clusters.

6) Anomaly Detection: Isolation Forest and Local Outlier Factor (LOF) were used to identify customers with unusual purchasing behavior.Both methods were compared to evaluate consistency between different anomaly detection approaches.

III. Results

A. Association Rule Mining

Association rule mining identified several meaningful product relationships.

Index	antecedents	consequents	confidence	lift
27	(ROSES REGENCY TEACUP AND SAUCER	(GREEN REGENCY TEACUP AND SAUCER)	0.894495	23.989564
29	(PINK REGENCY TEACUP AND SAUCER, GI	(ROSES REGENCY TEACUP AND SAUCER)	0.847826	20.0663
7	(PINK REGENCY TEACUP AND SAUCER)	(GREEN REGENCY TEACUP AND SAUCER)	0.827338	22.188466
22	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.784173	18.559754
11	(GREEN REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER)	0.782923	18.530185
6	(GARDENERS KNEELING PAD CUP OF TEA	(GARDENERS KNEELING PAD KEEP CALM)	0.729134	17.873424
28	(ROSES REGENCY TEACUP AND SAUCER	(PINK REGENCY TEACUP AND SAUCER)	0.720887	24.027846
30	(PINK REGENCY TEACUP AND SAUCER)	(ROSES REGENCY TEACUP AND SAUCER , (	0.701439	24.027846
10	(ROSES REGENCY TEACUP AND SAUCER	(GREEN REGENCY TEACUP AND SAUCER)	0.690932	18.530185
4	(DOLLY GIRL LUNCH BOX)	(SPACEBOY LUNCH BOX)	0.688312	18.119023

Fig. 1. Association rule mining results

Tea cup and gift-related products frequently appeared together in transactions. Many rules showed high lift values, suggesting strong product relationships rather than random cooccurrence.

These findings can support recommendation systems, cross-selling strategies, and product bundling.

B. Customer Segmentation

Customer segmentation was performed using K-Means clustering based on the RFM features: Recency, Frequency, and Monetary value. The clustering analysis identified four major customer groups with distinct purchasing behaviors.

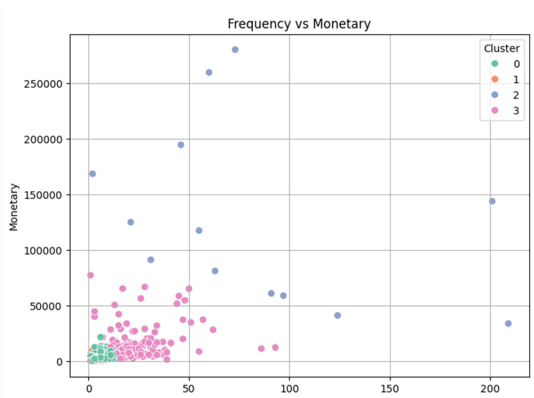

Fig. 2. Frequency versus Monetary scatter plot showing customer distribution across clusters

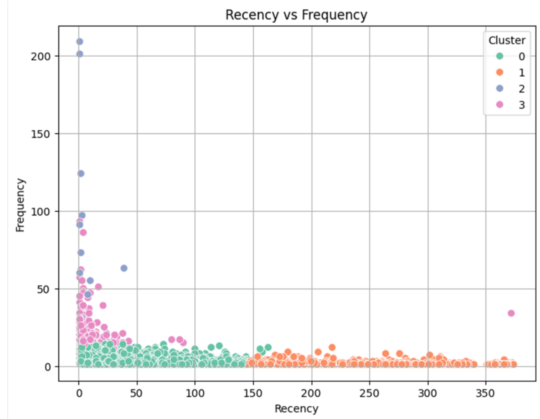

Fig. 3. Recency versus Frequency scatter plot illustrating customer engagement patterns.

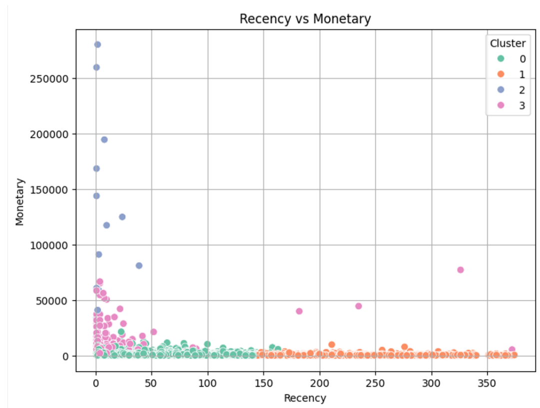

Fig. 4. Recency versus Monetary scatter plot showing differences in customer activity and spending behavior.

Cluster	CustomerID	Recency	Frequency	Monetary	Cluster_PCA
0	15289.12835625409 4	43.7026850032744	3.682711198428291	1359.0492838899802	0.002619515389652914 3
1	15347.79194001874 4	248.0759137769447	1.5520149953139644	480.6174798500469	0.9981255857544518
2	15435.0	7.384615384615385	82.53846153846153	127338.31384615385	2.076923076923077
3	15212.857843137255	15.5	22.333333333333332	12709.090490196078	2.4705882352941178

Fig. 5. Cluster summary table showing average RFM values and customer characteristics for each cluster.

The clustering analysis identified:

- Cluster 2: VIP or elite customers
- Cluster 3: Loyal high-value customers
- Cluster 0: Regular customers
- Cluster 1: Inactive customers

Cluster 2 showed the highest-value customer group in the dataset. These customers have an average Recency of approximately 7 days, an average Frequency of about 82 purchases, and an average Monetary value of around £127,338. Compared with all other clusters, this group shows substantially higher spending and purchasing activity. These customers may represent VIP customers, wholesale buyers, or corporate accounts that contribute a large portion of overall revenue.

Cluster 3 customers also demonstrate relatively strong customer loyalty and higher spending behavior, although their purchasing activity is less extreme than Cluster 2. In contrast, Cluster 1 customers show low purchasing frequency, low spending levels, and long inactivity periods, suggesting potential customer churn risk.

Overall, the clustering analysis identified clear differences between high-value customers, loyal customers, regular customers, and inactive customer groups.

C. PCA Visualization

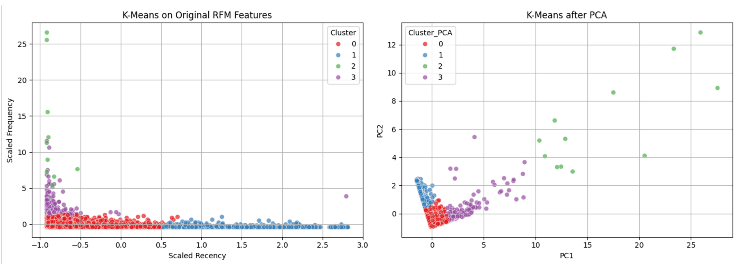

Fig. 6. PCA visualization of customer clusters

The PCA visualization shows that customer clusters become more compact and easier to distinguish after dimensionality reduction. Compared with clustering on the original RFM features, PCA improves cluster separation and provides a clearer overall view of customer structure. Although some overlap between clusters still exists, the customer groups become more visually interpretable after transformation.

The PCA results also support the clustering analysis by showing that customer groups follow identifiable behavioral patterns rather than completely random distributions.

D. Decision Tree Interpretation

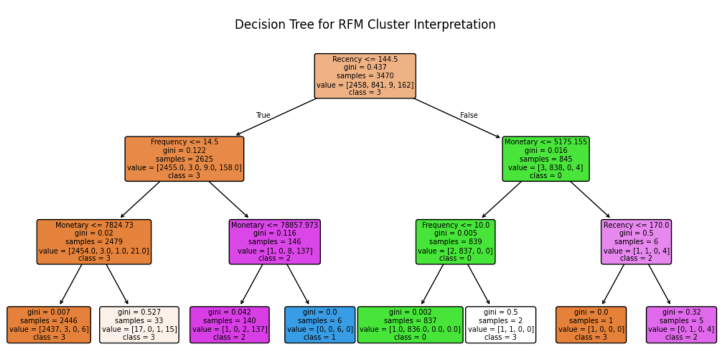

Fig. 7. Decision tree interpretation of customer cluster

The decision tree showed that Frequency and Monetary were the most important features for identifying valuable customer groups.

One important result showed that customers with recent purchases and high spending may represent wholesale buyers or company accounts.

Another result showed that some customers with historically high spending but long inactivity periods may represent dormant high-value customers.

E. Naive Bayes Analysis

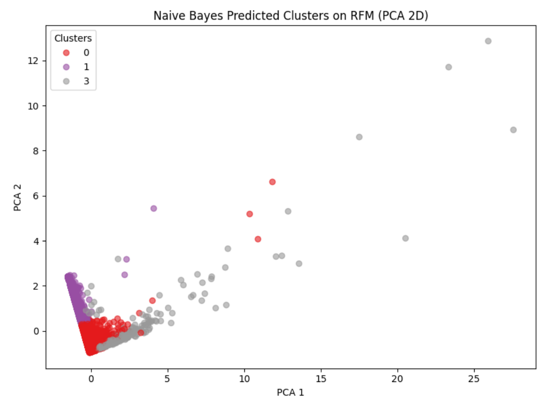

Fig. 8. Naive Bayes predicted clusters on PCA-transformed RFM data

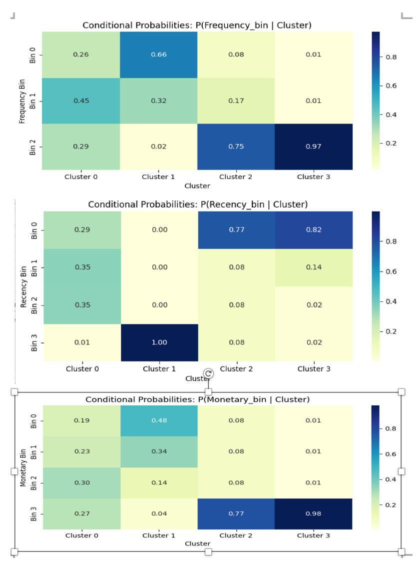

Fig. 9. conditional probabilities

The Naive Bayes analysis demonstrated strong probabilistic consistency between RFM features and customer cluster structure, with approximately 87% agreement across customer groups.

VIP customers were strongly associated with high Frequency and Monetary values, while inactive customers were associated with high Recency values.

The PCA visualization showed that customer groups could be reasonably separated, although some overlap still existed.

F. Anomaly Detection

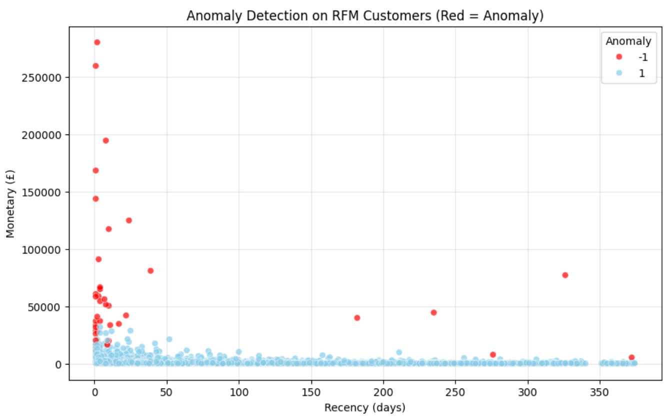

Fig. 10. Anomaly detection on RFM Customers

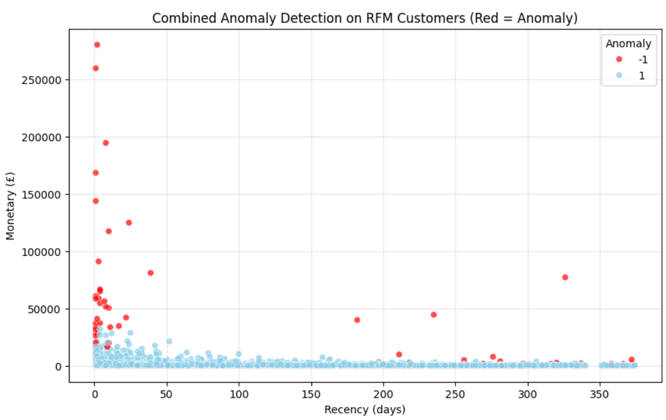

Fig. 11. Combined anomaly detection on RFM Customers

Both Isolation Forest and LOF identified similar anomalous customers with unusual spending or purchasing behavior.

For example, CustomerID 14911 showed a purchase frequency of 201 purchases, suggesting possible wholesale purchasing behavior.

An important analytical question is whether the VIP customer cluster and anomaly detection results represent the same customer population.

The results suggest that these groups are partially overlapping rather than identical.

Cluster 2 customers showed extremely high Frequency and Monetary values, with an average Monetary value of approximately £127,338.

Similarly, anomaly detection identified many customers with unusually high spending and purchasing frequency.

However, not all anomalies belonged to the VIP cluster. Some anomalies showed unusual purchasing timing or irregular customer behavior instead of stable high-value purchasing behavior.

This suggests that clustering and anomaly detection provide complementary insights rather than duplicate information.

IV. Interpretation

The findings provide several meaningful insights into customer behavior in online retail environments.

One important finding is that a very small group of VIP customers contributes a disproportionately large amount of total revenue. These customers may represent wholesalers, corporate buyers, or elite long-term customers.

Loyal customers also play an important role in maintaining long-term business profitability.

In contrast, inactive customers showed signs of customer churn risk due to low purchasing activity and long inactivity periods.

The association rule mining results support recommendation systems, cross-selling strategies, and product bundling.

Anomaly detection identified statistically unusual customers, including potential wholesalers and customers with irregular purchasing behavior.

Overall, the project demonstrates how multiple data mining techniques can generate actionable business insights for customer retention, marketing strategies, recommendation systems, and customer analytics.

V. Critical Assessment

A. Validity

Several factors suggest that the findings are reasonably reliable.

Multiple methods identified similar customer behavior patterns. Clustering, decision tree interpretation, Naive Bayes analysis, and anomaly detection consistently highlighted the importance of Frequency and Monetary features.

PCA also improved cluster separation and visualization.

In addition, K-Means clustering was tested with multiple random seeds and produced generally consistent cluster structures.

These overlapping findings increase confidence that the discovered patterns are meaningful rather than random noise.

These results suggest meaningful customer behavior patterns in this dataset, but they should not be interpreted as universal retail customer categories without additional validation on other datasets.

B. Limitations

This project also has several limitations.

First, the dataset is heavily dominated by UK customers, which limits the generalizability of the findings.

Second, the dataset contains strong class imbalance because VIP customers represent only a small portion of the customer population.

Another limitation is parameter sensitivity. Different parameter choices for K-Means, Isolation Forest, and LOF may produce different clustering or anomaly detection results.

The preprocessing stage may also introduce bias because transactions with missing CustomerID values were removed.

Finally, the project mainly focuses on transaction-based behavioral analysis using RFM features and does not include demographic or seasonal information.

C. Ethical Considerations

Although the dataset is anonymized, customer behavior analysis still raises important ethical considerations.

Customer privacy is an important concern in retail analytics. Businesses should ensure that customer data is used responsibly and securely.

Bias awareness is also important because the dataset mainly represents UK customers.

In addition, businesses should avoid focusing only on high-value customers while ignoring low-frequency customers who may still have future business potential.

VI. Conclusion

This project explored customer behavior patterns in the UCI Online Retail dataset using multiple data mining techniques.

Association rule mining revealed meaningful product relationships that may support recommendation systems and cross-selling.

Customer segmentation identified VIP customers, loyal customers, regular customers, and inactive customers.

Decision tree interpretation and Naive Bayes analysis provided additional insight into customer purchasing behavior.

Anomaly detection identified unusual customers with extremely high spending or irregular purchasing behavior.

The project also showed that VIP customers and anomalies are partially overlapping rather than identical populations.

Overall, the project demonstrates how multiple data mining techniques can work together to better understand customer behavior and support business decision-making.

Future work could include:

- DBSCAN comparison
- FP-Growth comparison
- hierarchical clustering
- time-series customer analysis
- stronger validation methods
- improved anomaly detection tuning

Through this project, important experience was gained in customer analytics, feature engineering, probabilistic interpretation, and anomaly detection.

VII. References

References

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